

# BANK MARKETING DATA ANALYSIS PROJECT

## Exploratory Data Analysis (EDA) using Python

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### Introduction

#### 1.1 Project Overview

This project focuses on performing **Exploratory Data Analysis (EDA)** on a bank marketing dataset to understand customer behaviour and analyse factors influencing subscription to a term deposit.

The analysis was performed using **Python in Google Colaba**, following a structured and systematic data analysis approach.

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#### 1.2 Objectives

- To load and inspect the dataset
  - To understand the structure and types of variables
  - To check missing values and data consistency
  - To analyse statistical properties of numerical features
  - To study customer subscription distribution
  - To identify correlations between numerical variables
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### Tools & Technologies Used

- **Programming Language:** Python
  - **Platform:** Google Colaba
  - **Libraries Used:**
    - Pandas (data handling)
    - Matplotlib (visualization)
    - Seaborn (advanced plots)
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### Dataset Description

- **Dataset Name:** bankmarketing.csv

- **Number of Records:** 41,188
- **Number of Features:** 21
- **Target Variable:** y
  - yes → Client subscribed to term deposit
  - no → Client did not subscribe

The dataset includes:

- Demographic information
- Campaign interaction details
- Financial status
- Economic indicators

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## Step-by-Step Analysis with Screenshots

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### 4.1 Data Loading

The dataset was loaded using the Pandas library.

#### Code Used:

```
import pandas as pd

df = pd.read_csv('bankmarketing.csv')
```

 **Screenshot 1:** Dataset loaded in Google Colaba  
(Insert screenshot showing dataset loaded)

The screenshot shows a Google Colab notebook titled "Bank\_Marketing\_Inspection\_test.ipynb". The code cell contains the following code:

```
# Display the first few rows
df.head()
```

The output of the code cell shows the first five rows of the dataset:

|   | age | job       | marital | education   | default | housing | loan | contact   | month | day |
|---|-----|-----------|---------|-------------|---------|---------|------|-----------|-------|-----|
| 0 | 56  | housemaid | married | basic.4y    | no      | no      | no   | telephone | may   |     |
| 1 | 57  | services  | married | high.school | unknown | no      | no   | telephone | may   |     |
| 2 | 37  | services  | married | high.school | no      | yes     | no   | telephone | may   |     |
| 3 | 40  | admin.    | married | basic.6y    | no      | no      | no   | telephone | may   |     |
| 4 | 56  | services  | married | high.school | no      | no      | yes  | telephone | may   |     |

The variable explorer on the right shows the dataset structure:

| age | job         | marital | education           | default | housing |
|-----|-------------|---------|---------------------|---------|---------|
| 56  | housemaid   | married | basic.4y            | no      | no      |
| 57  | services    | married | high.school         | unknown | no      |
| 37  | services    | married | high.school         | no      | yes     |
| 40  | admin.      | married | basic.6y            | no      | no      |
| 56  | services    | married | high.school         | no      | no      |
| 45  | services    | married | basic.9y            | unknown | no      |
| 59  | admin.      | married | professional.course | no      | no      |
| 41  | blue-collar | married | unknown             | unknown | no      |
| 24  | technician  | single  | professional.course | no      | yes     |
| 25  | services    | single  | high.school         | no      | yes     |

✓ The dataset loaded successfully without errors.

## 4.2 Displaying Initial Records

To understand the structure of the data, the first five rows were displayed.

### Code Used:

```
df.head()
```

**Screenshot 2:** First five rows of the dataset

Bank\_Marketing\_Inspection\_test.ipynb

```
import pandas as pd

# Load the dataset
df = pd.read_csv('bankmarketing.csv')

# Display the first few rows
df.head()
```

|   | age | job       | marital | education   | default | housing | loan | contact   | month | day |
|---|-----|-----------|---------|-------------|---------|---------|------|-----------|-------|-----|
| 0 | 56  | housemaid | married | basic.4y    | no      | no      | no   | telephone | may   |     |
| 1 | 57  | services  | married | high.school | unknown | no      | no   | telephone | may   |     |
| 2 | 37  | services  | married | high.school | no      | yes     | no   | telephone | may   |     |
| 3 | 40  | admin.    | married | basic.6y    | no      | no      | no   | telephone | may   |     |
| 4 | 56  | services  | married | high.school | no      | no      | yes  | telephone | may   |     |

5 rows x 21 columns

bankmarketing.csv

1 to 10 of 4118 entries

| age | job         | marital | education           | default | housing |
|-----|-------------|---------|---------------------|---------|---------|
| 56  | housemaid   | married | basic.4y            | no      | no      |
| 57  | services    | married | high.school         | unknown | no      |
| 37  | services    | married | high.school         | no      | yes     |
| 40  | admin.      | married | basic.6y            | no      | no      |
| 56  | services    | married | high.school         | no      | no      |
| 45  | services    | married | basic.9y            | unknown | no      |
| 59  | admin.      | married | professional.course | no      | no      |
| 41  | blue-collar | married | unknown             | unknown | no      |
| 24  | technician  | single  | professional.course | no      | yes     |
| 25  | services    | single  | high.school         | no      | yes     |

Show 10 per page

1 2 10 100 1000 4000 4100 4110 4119

### Observation:

- The dataset contains both numerical and categorical variables
- Columns such as age, job, marital, education, etc. are visible

### 4.3 Checking Data Types & Missing Values

To ensure data quality, missing values and data types were checked.

#### Code Used:

```
df.info()
```

```
df.isnull().sum()
```

**Screenshot 3:** Data types and missing values output

The screenshot shows a Google Colab notebook titled 'Bank\_Marketing\_Inspection\_test.ipynb'. The code cell contains the following Python code:

```
# Check for missing values and data types

df.info()
df.isnull().sum()
```

The output of the code shows the following data types and counts:

| Column         | Count | Non-null | Data Type |
|----------------|-------|----------|-----------|
| month          | 41188 | non-null | object    |
| day_of_week    | 41188 | non-null | object    |
| duration       | 41188 | non-null | int64     |
| campaign       | 41188 | non-null | int64     |
| pdays          | 41188 | non-null | int64     |
| previous       | 41188 | non-null | int64     |
| poutcome       | 41188 | non-null | object    |
| emp.var.rate   | 41188 | non-null | float64   |
| cons.price.idx | 41188 | non-null | float64   |
| cons.conf.idx  | 41188 | non-null | float64   |
| euribor3m      | 41188 | non-null | float64   |
| nr.employed    | 41188 | non-null | float64   |
| y              | 41188 | non-null | object    |

The variable inspector on the right shows the first 10 rows of the 'bankmarketing.csv' dataset:

| age | job         | marital | education           | default | housing |
|-----|-------------|---------|---------------------|---------|---------|
| 56  | housemaid   | married | basic.4y            | no      | no      |
| 57  | services    | married | high.school         | unknown | no      |
| 37  | services    | married | high.school         | no      | yes     |
| 40  | admin.      | married | basic.6y            | no      | no      |
| 56  | services    | married | high.school         | no      | no      |
| 45  | services    | married | basic.9y            | unknown | no      |
| 59  | admin.      | married | professional.course | no      | no      |
| 41  | blue-collar | married | unknown             | unknown | no      |
| 24  | technician  | single  | professional.course | no      | yes     |
| 25  | services    | single  | high.school         | no      | yes     |

### Observation:

- No missing values found in any column
- Dataset is clean and ready for analysis
- Mixture of int64, float64, and object data types

✓ No preprocessing for missing values was required.

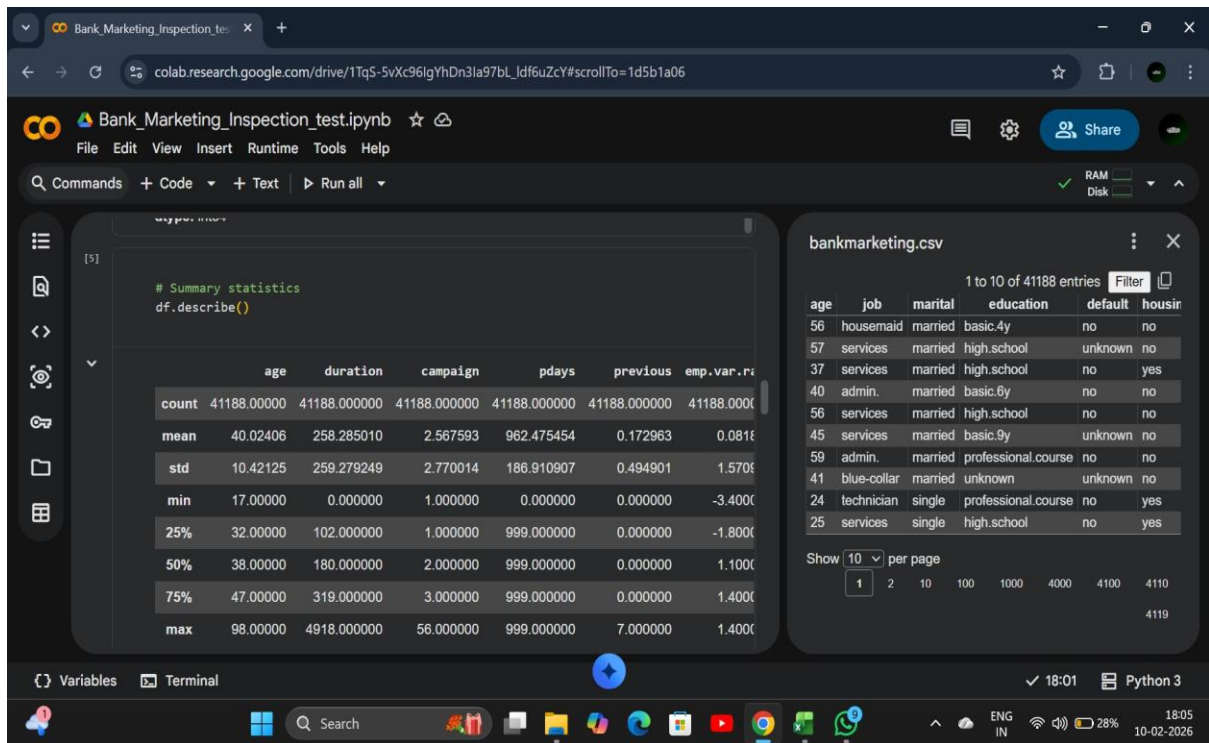
## 4.4 Summary Statistics of Numerical Features

Descriptive statistics were generated to understand numerical data distribution.

### Code Used:

```
df.describe()
```

 **Screenshot 4:** Summary statistics table



## Key Insights:

- Average customer age  $\approx$  **40 years**
- Most campaigns contacted clients **1–3 times**
- duration has a wide range, indicating its strong influence
- Economic indicators vary significantly

## 4.5 Target Variable Distribution

To analyze customer subscription behavior, the target variable was visualized.

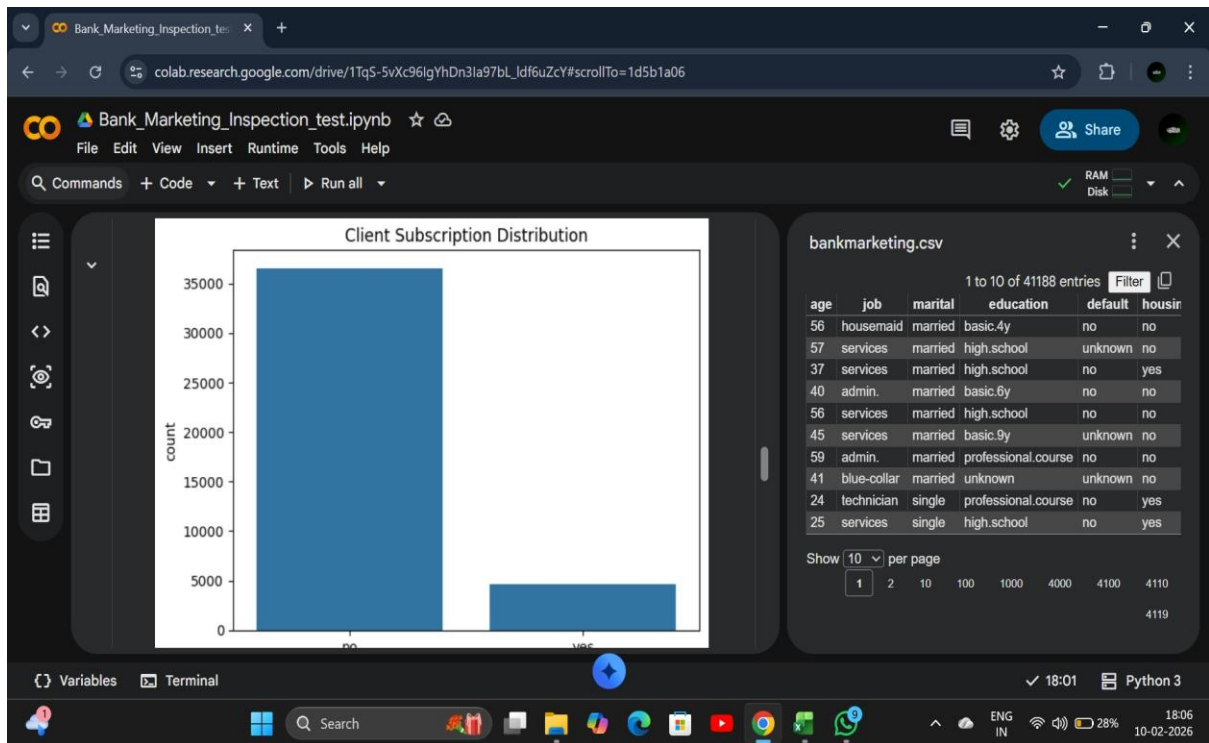
### Code Used:

```
sns.countplot(x='y', data=df)
```

```
plt.title("Client Subscription Distribution")
```

```
plt.show()
```

 **Screenshot 5:** Client subscription distribution bar chart



### Observation:

- Majority of clients **did not subscribe**
- Dataset is **imbalanced**
- Important insight for future machine learning models

## 4.6 Correlation Analysis of Numerical Features

Correlation analysis was performed to identify relationships between numerical variables.

### Code Used:

```
numeric_df = df.select_dtypes(include=['int64', 'float64'])
```

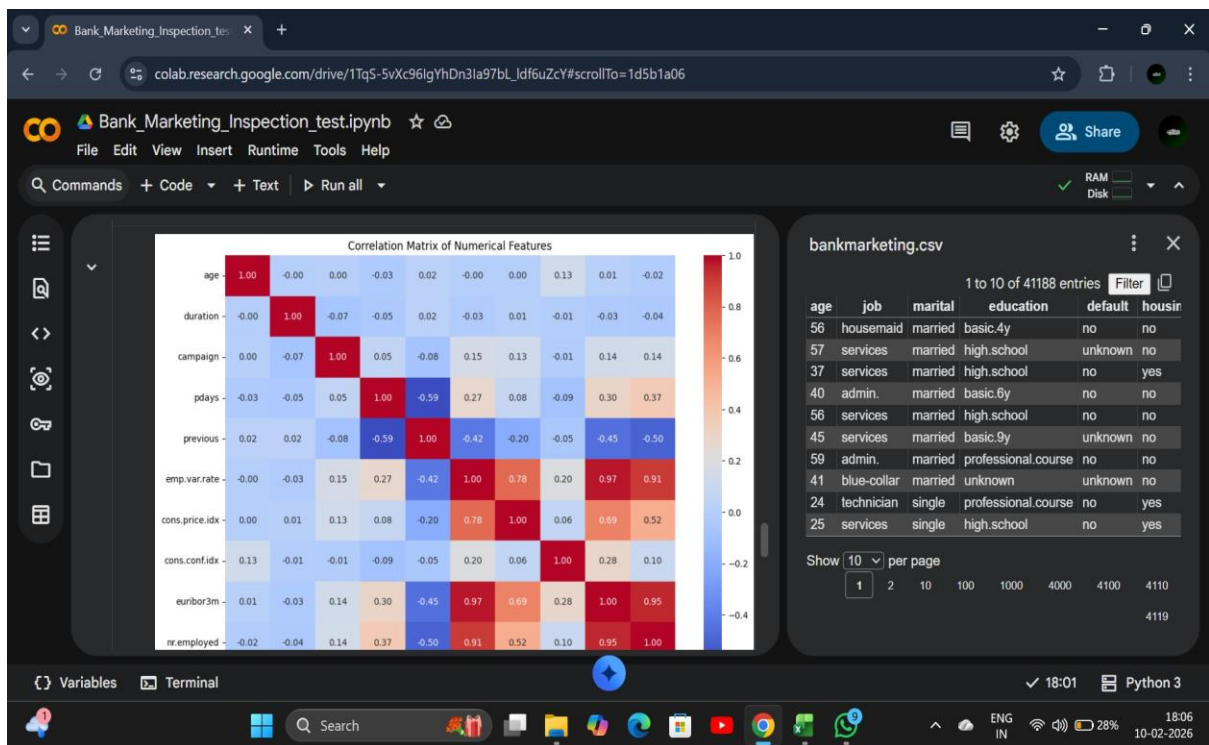
```
corr_matrix = numeric_df.corr()
```

```
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
```

```
plt.title("Correlation Matrix of Numerical Features")
```

```
plt.show()
```

 **Screenshot 6:** Correlation heatmap



## Key Findings:

- Strong positive correlation between:
  - emp.var.rate, euribor3m, nr.employed
- Negative correlation between:
  - pdays and previous
- Demographic variables show weak correlation

✓ Economic indicators strongly influence customer behavior.

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## 5 Key Insights Summary

- Dataset is **clean and well-structured**
- Customer subscription rate is low
- Campaign duration plays a crucial role
- Economic factors influence outcomes more than demographics
- Dataset is suitable for predictive modeling

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## 6 Conclusion



This exploratory data analysis provided a clear understanding of the bank marketing dataset. The analysis identified key patterns in customer behavior and campaign effectiveness. The dataset is suitable for further preprocessing and machine learning model development to predict customer subscription outcomes.

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## **7 Future Scope**

- Encode categorical variables
- Handle class imbalance
- Train classification models
- Evaluate model performance
- Improve marketing strategy effectiveness